# MATHEMATICAL GAME THEORY

Final Examination

Fall 2024 · Prof. L. Nakamura

180 minutes · 100 points

Name: _____ ID: _____

**Instructions.** Two sheets of handwritten notes are allowed; no other materials or devices. There are six problems. Proofs should be complete and concise; cite named theorems from lecture where you use them. You may use the result of an earlier part in a later part even if you did not prove it. Leave numerical answers as exact fractions.

| Problem | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points | 16 | 18 | 18 | 20 | 14 | 14 | 100 |
| Score | | | | | | | |

**Problem 1** *Correlated equilibrium* (16 points)

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Two drivers approach a one-lane bridge; each chooses Dare (D) or Chicken (C):

| | D | C |
|---|---|---|
| D | 0, 0 | 8, 2 |
| C | 2, 8 | 6, 6 |

A distribution $z = (z_{\mathsf{DD}}, z_{\mathsf{DC}}, z_{\mathsf{CD}}, z_{\mathsf{CC}})$ on action profiles is a *correlated equilibrium* (CE) if no player, told only her own recommended action, gains in expectation by deviating from it.

(a) [3] Find all Nash equilibria of the game and their payoffs.

(b) [5] Write the CE incentive constraints and show that they reduce to four linear inequalities of the form $z_\alpha \geq z_\beta$.

(c) [5] Find the CE that maximizes the sum of the players' expected payoffs. Prove that it is optimal and unique, and compare the players' payoffs with those in the Nash equilibria.

(d) [3] Prove that for any finite two-player game, the product distribution $z(s_1, s_2) = \sigma_1(s_1)\sigma_2(s_2)$ induced by a mixed Nash equilibrium $(\sigma_1, \sigma_2)$ is a correlated equilibrium.

**Problem 2** *Shapley value and the core* (18 points)

A transferable-utility game on the player set $N$ is a function $v : 2^N \to \mathbb{R}$ with $v(\varnothing) = 0$. For an ordering $\pi$ of $N$, let $P_i^\pi$ be the set of players preceding $i$ in $\pi$. The Shapley value is

$$\phi_i(v) = \frac{1}{|N|!} \sum_{\pi} \Big[ v\big(P_i^\pi \cup \{i\}\big) - v\big(P_i^\pi\big) \Big].$$

(a) [5] Prove that $\sum_{i \in N} \phi_i(v) = v(N)$ (efficiency), and that if $v(S \cup \{i\}) = v(S)$ for every $S \subseteq N \setminus \{i\}$ then $\phi_i(v) = 0$ (null player).

(b) [5] Let $N = \{1, 2, 3\}$ with $v(\{i\}) = 0$, $v(\{1,2\}) = 8$, $v(\{1,3\}) = 6$, $v(\{2,3\}) = 4$, and $v(N) = 10$. Compute the Shapley value.

(c) [5] Describe the core of this game, show that it is nonempty by listing its extreme points, and decide whether the Shapley value lies in the core.

(d) [3] In the weighted voting game $[5; 3, 2, 2]$ (a coalition wins iff its total weight is at least 5), compute the Shapley–Shubik power index of each player.

**Problem 3** *Evolutionary stability* (18 points)

In a symmetric two-player game with payoff $u(\sigma, \tau)$ to a player using $\sigma$ against $\tau$, a strategy $\sigma$ is an *evolutionarily stable strategy* (ESS) if for every $\tau \neq \sigma$ either (i) $u(\sigma, \sigma) > u(\tau, \sigma)$, or (ii) $u(\sigma, \sigma) = u(\tau, \sigma)$ and $u(\sigma, \tau) > u(\tau, \tau)$.

(a) [4] Prove that if $(\sigma, \sigma)$ is a *strict* Nash equilibrium, i.e. $u(\sigma, \sigma) > u(\tau, \sigma)$ for all $\tau \neq \sigma$, then $\sigma$ is an ESS; and that every ESS $\sigma$ gives a symmetric Nash equilibrium $(\sigma, \sigma)$.

(b) [8] Consider Hawk–Dove with resource value $V = 6$ and fighting cost $K = 10$:

$$u(H, H) = \tfrac{V - K}{2} = -2, \quad u(H, D) = 6, \quad u(D, H) = 0, \quad u(D, D) = 3.$$

Show that neither pure strategy is an ESS. Find the symmetric mixed equilibrium $p^* = \Pr(H)$ and prove that it is an ESS by computing $u(p^*, q) - u(q, q)$ for an arbitrary mixed strategy $q$.

(c) [6] Let $x(t)$ be the population share of Hawks, evolving by the replicator dynamic $\dot{x} = x\,[u(H, x) - \bar{u}(x)]$ with $\bar{u}(x) = u(x, x)$. Show that $\dot{x} = x(1 - x)(3 - 5x)$, find all rest points, and prove that $x(t) \to p^*$ for every initial condition $x(0) \in (0, 1)$.

**Problem 4** *Stable matching* (20 points)

Four doctors $d_1, \dots, d_4$ are to be matched one-to-one with four hospitals $h_1, \dots, h_4$ (one position each). Preferences (most preferred first):

| | | | |
|---|---|---|---|
| $d_1$: | $h_2 \succ h_1 \succ h_3 \succ h_4$ | $h_1$: | $d_3 \succ d_4 \succ d_1 \succ d_2$ |
| $d_2$: | $h_1 \succ h_2 \succ h_4 \succ h_3$ | $h_2$: | $d_4 \succ d_3 \succ d_2 \succ d_1$ |
| $d_3$: | $h_4 \succ h_2 \succ h_3 \succ h_1$ | $h_3$: | $d_1 \succ d_3 \succ d_4 \succ d_2$ |
| $d_4$: | $h_4 \succ h_3 \succ h_2 \succ h_1$ | $h_4$: | $d_2 \succ d_4 \succ d_1 \succ d_3$ |

(a) [6] Run the doctor-proposing deferred acceptance algorithm, showing the proposals and rejections in each round. State the resulting matching.

(b) [6] Prove that for any preferences (with equal numbers of doctors and hospitals, all acceptable), the matching produced by doctor-proposing deferred acceptance is stable.

(c) [4] Run the hospital-proposing algorithm. Using the fact that every doctor weakly prefers the doctor-proposing outcome to any stable matching, and weakly prefers any stable matching to the hospital-proposing outcome, determine which doctors have the same partner in every stable matching.

(d) [4] In the *roommates problem*, four students $A, B, C, D$ must be split into two pairs. Preferences over roommates are $A : B \succ C \succ D$, $B : C \succ A \succ D$, $C : A \succ B \succ D$, and $D$ is arbitrary. Prove that no pairing is stable (i.e. for every pairing there are two students who prefer each other to their assigned roommates).

**Problem 5** *First-price auctions* (14 points)

Bidders' values are independent with CDF $F(v) = v^2$ on $[0, 1]$. The highest bid wins; ties occur with probability zero in what follows.

(a) [6] With two bidders, show that $b(v) = \frac{2}{3}v$ is a symmetric Bayesian Nash equilibrium of the first-price auction. Your argument must establish that the equilibrium bid is a *global* best response, including against bids outside the range of $b$.

(b) [4] Compute the seller's expected revenue in this equilibrium, and in the second-price auction with truthful bidding. Comment.

(c) [4] With $n \geq 2$ bidders, show that $b_n(v) = \dfrac{2n-2}{2n-1}\, v$ is a symmetric Bayesian Nash equilibrium.

**Problem 6** *Guessing two-thirds of the average* (14 points)

Each of $n \geq 2$ players chooses $x_i \in [0, 100]$. Let $m_{-i} = \frac{1}{n-1}\sum_{j \neq i} x_j$ be the average of the others' choices. Player $i$'s payoff is

$$u_i(x) = -\Big(x_i - \tfrac{2}{3}\, m_{-i}\Big)^2.$$

(a) [4] Find player $i$'s best response and prove that every $x_i > \frac{200}{3}$ is strictly dominated.

(b) [6] Prove by induction that, for every $k \geq 0$, the set of strategies of each player surviving $k$ rounds of iterated elimination of strictly dominated strategies is exactly $\big[0, (\tfrac{2}{3})^k \cdot 100\big]$. Conclude that only $x_i = 0$ survives.

(c) [4] Give a direct proof, not using part (b), that $(0, \dots, 0)$ is the unique Nash equilibrium.

END OF EXAMINATION